

What is Conversational AI

- Conversational artificial intelligence (AI) is technology that enables humans and computers to communicate clearly and effectively through speech or text.
- Conversational AI works by recognizing a user's speech or text patterns, predicting their intent and responding with an adaptive, automated script.
- The best conversational AI platforms today can create truly natural, human-like interactions.

Conversational AI examples in use today

- There are many examples of conversational AI applications currently being used across all industries.
- From customer service to marketing to security, these highly intelligent programs are helping businesses connect with customers and employees in never-before-seen ways.
- In fact, conversational AI has become a centerpiece of many organizations' digital transformation and [cloud migrations](#) in the wake of the global pandemic.

Examples of conversational AI currently in use include:

- Voice assistants, like Amazon's Alexa and Apple's Siri that interact via voice AI
- Virtual assistants, such as Microsoft's Cortana
- AI powered chatbots
- Future conversational AI applications already in development include home automation, automobiles, marketing, advertising and e-learning.
- As our exposure to and appetite for interactive technology increases, conversational AI will play a bigger and bigger role in our everyday lives.

The technology behind conversational AI

- Conversational AI as we know it is driven by three technological developments
- **Natural Language Processing**
 - Natural language processing (NLP) software [analyzes natural human language and speech](#), interpreting contextual nuances and extracting relevant information. Together with natural language understanding (NLU), NLP is what allows humans to have conversations with artificial intelligence.
- **Artificial Intelligence**
 - Artificial intelligence (AI) uses the data analyzed by NLP to predict patterns of communication. Conversational artificial intelligence is what allows machines to communicate back and forth with humans, generating relevant automated responses based on the speaker's intent and other contextual insights.
- **Machine Learning**
 - Lastly, machine learning (ML) enables AI-based systems to “learn” and improve from experience without being explicitly programmed. A subset of machine learning, deep learning, allows conversational AI models to cluster and categorize extracted data to make highly accurate predictions. Deep learning models run on neural networks of three or more layers, allowing them to perform end-to-end learning automatically and scale their algorithms with data. Many voice AI-based virtual assistants use deep learning models to mimic natural human speech.
 - Today, natural language processing, artificial intelligence and machine learning power a range of applications, including search engines, email spam filters, language translation software, voice assistant, grammar analysis and even social media management applications.

Conversational AI vs Chatbot-

Conversational AI vs Chatbot- What is the Difference between?

Chatbot	Conversational AI
lacks capability beyond performing a limited number of functions	No such limits in performing various functions across sales, support and marketing
Can best answer basic FAQs	Can make chatbots smarter
Unable to interpret human language	Able to interpret human language
Text-only command	Both voice and Text commands
The consistent effort is towards adding more channels for communication	Ideal for omnichannel approach
Enables scripted conversational flow	Has natural language processing ability
Supports linear interactions	Supports dynamic interactions

Conversational AI vs Chatbot-

- There are more differences between a chatbot and conversational AI and some of them include –
- **Limitations in functionality** – A conventional chatbot lacks capability beyond performing a limited number of functions. In most cases, its ability stretches to answering basic FAQs. On the other hand, conversational AI is a technology that can boost a chatbot's ability manifold and make it easily understand and interpret human language.
- **Text-only command vs Voice and text commands** – While a chatbot is capable of text-only commands (inputs & outputs), conversational AI can go a step further and enable both, voice as well as text commands at the same time.
- **Single channel vs Omnichannel** – Most chatbots are limited to being used as a chat interface only and this makes it fit for a single channel alone. On the other hand, conversational AI can enable [omnichannel communication](#) by being deployed across websites, voice assistants, etc.
- **Scripted vs natural language processing** – Since chatbot is basic in technology it can only enable scripted conversational flow based on pre-defined rules. And when there is conversational AI, it can come with the ability of natural language processing, therefore being easily able to understand and contextualize conversations.
- **Linear vs dynamic interactions** – A chatbot can best support linear interactions whereas a conversational AI can go steps up and enable non-linear and dynamic interactions. For that reason, AI-enabled bots are best suited for [customer support](#) tasks.
- **Navigation vs Dialogue** – Compared to chatbots which support navigation, conversational AI is obviously more advanced and can support the dialogue of any complexity.

Benefits of conversational AI

- **Increase customer satisfaction and engagement with fast and interactive responses**
- Customers want immediate service, and according to the latest [Zendesk Customer Experience Trends Report](#), 71 percent of them believe AI and chatbots help them get faster replies. By using chatbots, your messaging channels can provide quick, convenient, [24/7 customer support](#).
- Conversational bots can also use rich messaging types—like carousels, quick replies, and embedded apps—to make [customer self-service](#) easier and enhance customer interactions.
- Since implementing a Zendesk chatbot, [Accor Plus](#) has seen a **20 percent increase** in customer satisfaction, a **352 percent increase** in response time, and a **220 percent increase** in resolution time. The bot provides around-the-clock support and offers self-service options to customers outside of regular business hours.
- “Automated and asynchronous messaging creates a more efficient mindset. Using Zendesk macros empowers agents to work on multiple channels at once while waiting for responses.” *Christopher Douglas, Director of Member Services for Accor Plus*
- **Access and use data about consumers to provide personalized responses**
- With conversational AI, you can tailor interactions based on each customer’s account information, actions, behavior, and more. The more tools you connect to your bot, the more data it has for personalization.
- Here are a few ways [bots can add personalization to the customer experience](#):
 - Asking shoppers if they have questions about items in their cart
 - Recommending products based on order history and personal information on the account
 - Providing the status of an order without asking for the order information
- [Fútbol Emotion](#) teamed up with Zendesk to implement a chatbot that used customer data to personalize the customer experience.
- “If a customer wants to know the status of an order and starts a conversation with our support team, the AI bot knows who they are without asking for the order number because that metadata is already stored in the system,” says Daniel Hsu, Ecommerce Manager at Fútbol Emotion

Benefits of conversational AI

- **Deliver proactive support to increase sales**
- It's nice to get a warm welcome when returning to a website. Chatbots are great at greeting customers, but that's not all they can do. Chatbots can understand the customer's buying habits and may proactively ask them if they'd like to get in touch with sales. They can offer self-service options based on prompts and understand when a customer might want a human agent to help them.
- **Improve agent efficiency and workflows**
- 74 percent of consumers think AI improves customer service efficiency, and they're right. A tool like [Zendesk bots](#) can respond to customers' simple, low-priority questions and lead them to a speedy resolution. Each support ticket a conversational AI chatbot can resolve is one less ticket your agents need to worry about.
- "Conversational AI is primarily used to drive customer satisfaction, but it ends up improving agent satisfaction as well." *Paul Lalonde, a Zendesk product expert*
- "One of the major reasons for call center turnover is that agents find tasks boring and repetitive," says Lalonde. But when you have bots working towards solving customer problems, agents don't have to worry about doing the more mind-numbing, repetitive tasks. They can just get involved when more critical thinking is needed."
- Conversational AI chatbots can also:
 - Identify high-value customers to prioritize their requests
 - Intelligently route tickets to the agent best suited to handle a request
 - Provide the customer with self-service options so they can resolve the request without the help of a human agent
 - The technology can relay relevant information when there's a bot-to-human handoff, too, giving agents the context they need to provide better support.
- **Gain a greater understanding of customer sentiment**
- AI chatbots can even help agents understand [customer sentiment](#), so the agent receiving the handoff knows how to tailor the interaction. With the Intelligent Triage feature, Zendesk uses AI to add valuable information to support tickets, such as customer intent, sentiment, and language predictions. The agent-facing AI application, [Smart Assist](#), acts as a co-pilot to help guide the agent through the conversation by providing extra context and suggestions.
- When Noom launched [Noom Mood](#), the company [asked Zendesk to implement AI](#) to analyze customer conversations, tickets, issues, and, most importantly, customer sentiment. These insights allowed Noom to create an educational campaign that improved customer sentiment and increased engagement with the app.

Types of conversational AI technology

Types of conversational AI technology



Chatbots

Computer programs that simulate human interaction and conversations



Voice assistants

AI applications that understand voice commands and complete tasks



Interactive voice response

Automated phone systems that reply to voice and keypad commands

Chatbots

- Chatbots are **computer programs designed to simulate human conversations**. They help customers find quick answers around-the-clock or effectively route them to the best department to handle their inquiry. Chatbots are usually implemented through social media messengers or chat applications built into a website or mobile app. Here are some different types of chatbots:
- **Rule-based chatbots** work like a flowchart, with humans mapping out conversations based on predefined rules. They're dependable, they're easy to program, and they integrate into your preferred customer support channel.

However, rule-based chatbots lack AI, so they don't allow for as much personalization or flexibility. They are typically used to automate processes, like answering simple FAQs.

- **Deep learning conversational AI chatbots** can independently lead a conversation. NLP enables an AI bot to generate relevant answers by analyzing the conversation and trying to understand customer intent. Based on the intent, machine learning then formulates a response.

AI chatbots are more challenging to set up than rule-based chatbots but are much more versatile and able to answer more complex queries. Ecommerce websites often use AI chatbots to understand the shopper's intent when providing recommendations.

- **Hybrid chatbots**—like [Zendesk chatbots](#)—are rule-based and AI-equipped. For example, a hybrid chatbot on a healthcare site might use AI technology to understand a patient's issue and rule-based technology to offer medical instructions.

Customer service providers can use hybrid chatbots to answer common questions and identify when to bring in an agent to help resolve an issue.

Voice assistants

- Voice assistants are **AI applications programmed to understand voice commands and complete tasks** for the user based on those commands. Starting with speech recognition, human speech converts into machine-readable text, which voice assistants can process in the same way chatbots process data.
- You will often find voice assistants in search engines, smart speakers, and operating systems. [Alexa](#) and [Siri](#) are common examples of this technology. The main advantages of voice assistants are that:
- Customers can use them hands-free, which makes them popular options for people with disabilities
- They can recognize a wide variety of languages, just like chatbots

Interactive voice response (IVR)

- Interactive voice response (IVR) is an automated phone system tool that greets callers and provides them with menu options via voice or keypad inputs. Based on the caller's responses, the IVR can:
- Guide the caller through a series of prompts, ultimately routing them to the department or agent they're trying to reach
- Provide immediate answers to FAQs
- Offer self-service options
- **IVR functions as a hybrid of chatbots and standard voice assistants,** combining mapped-out conversations with a verbal interface.

Conversational AI examples and use cases

- Businesses can use conversational AI for their chatbots, voice assistants, and IVRs. But what is an example of using conversational AI in a business setting? Here are a few use cases to consider, with additional info below:
- Customer service
- Sales and marketing
- Data collection
- Internet of Things (IoT)

Conversational AI examples and use cases

- **Customer service**

- [Customer service teams use chatbots](#) in various ways. From greeting customers, offering around-the-clock support, providing self-service options, and offering personalized recommendations during the shopping experience, conversational AI software has proven to be a versatile tool for creating great customer experiences.
- [Upwork's](#) mighty team of 300 support agents handles over 600,000 tickets each year. How do they do it? With help from Zendesk, the company utilizes chatbots to offer proactive support and deflect tickets by offering customers self-service options—resulting in a 58 percent chatbot resolution rate. These implementations have taken both the customer and agent experience to the next level and improved Upwork's overall customer service.

- **Sales and marketing**

- Businesses can use conversational AI software in their sales and marketing strategy to convert leads and drive sales. They can use it to provide a shopping experience for the customer that allows them to have a “virtual sales agent” that answers questions or provides recommendations. Zendesk chatbots can surface help center articles or answer FAQs about products in a customer's cart to nudge the conversion, too.
- Through data collected during interactions, chatbots can provide valuable information to help market products and services and identify customer trends and behaviors.

Conversational AI examples and use cases

- **Data collection**

- Conversational AI bots can capture key customer information like their name, email address, order numbers, and previous questions or issues. They can even pass all this data to an agent during the handoff by automatically adding it to the open ticket. This provides the agent with the context of the inquiry, so the customer doesn't need to repeat information.
- Agents can use this information to personalize interactions as well. According to our CX Trends Report, 59 percent of consumers believe businesses should use the data they collect about them to personalize their experiences.

- **Internet of Things (IoT) devices**

- [Internet of Things \(IoT\) devices](#) are the everyday devices people use that connect to the internet. Mobile phones, tablets, and smartwatches are all examples of IoT devices. They contain sensors that send real-time data to the agent when a customer reaches out about an issue.
- IoT sensors can even be placed inside industrial equipment, machinery, or vehicles to collect performance data. AI then analyzes the information to find patterns and predict when a device might need maintenance. AI can then proactively send alerts to the user or customer support.

How to implement conversational AI

- Learn how to implement conversational AI so you can start reaping the rewards:
- Establish your goals and use case
- Get support from stakeholders
- Determine your budget and resources
- Consider your existing infrastructure
- Choose and connect your CRM
- Look at data to measure performance

- **1. Establish your goals and use case**

- You won't know if your conversational AI initiative is paying off unless you know what you want to gain by using the technology.
- Do you want [conversational AI for support](#), sales, or marketing? Be specific about your objectives and the problems you want to solve so you can gauge which conversational AI technology is best for your company.
- For example, say your primary pain point is that your support agents are wasting time answering basic questions, and you want them available to handle complex customer inquiries. What types of conversational AI would best solve this issue? Perhaps it's a combination of voice assistants that deliver automated answers to common questions and rule-based chatbots that can address FAQs.
- Specify what customer service goals and key performance indicators (KPIs) you want to achieve before moving forward with implementation. That way, you can measure the success of your conversational AI strategy once it's in place.

- **Get support from stakeholders**

- The next step is securing support for the initiative. When pitching your idea to stakeholders, make sure you closely align your arguments with top business objectives. Focus on the importance of:
- **Understanding customer needs:** Demonstrate how conversational AI tools learn about customer needs, behaviors, and preferences—and explain how that will improve CX.
- **Improving agent satisfaction:** Emphasize the positive impact AI can have on your agents. Spending less time on repetitive tasks increases both productivity and employee satisfaction.
- **Getting a good return on investment:** Decision-makers will want clear ROI projections. Use resources like [Dataiku](#) and [Nexocode](#) to learn how to calculate, frame, and pitch the ROI metrics of AI projects.
- The success of your conversational AI initiative hinges on the support it receives across your organization. According to Deloitte's [State of AI report](#), AI projects cannot succeed if company leaders aren't setting core, overarching business strategies to achieve the vision.

- **Determine your budget and resources**
- After deciding how you'd like to use your chatbot, consider how much money and resources your business can allocate. For businesses with a small dev team, a no-code option would be a great fit because it works right out of the box. More complex use cases require additional budget and resources.

- **Consider your existing infrastructure**
- Next, investigate your current communication channels and existing infrastructure. Pick a conversational AI tool that can easily integrate with your current [customer support or sales CRM](#). You'll want the bot to work with the channels you already have and seamlessly step into current conversations for a great [omnichannel experience](#).

- **Choose and connect your customer support or sales CRM**
- Determine any additional tools you may require. What conversational AI platform investments have you already made (if any)? Leverage any existing architecture to deliver value and reduce costs. Does it integrate with your current systems?
- For example, if you already have a messenger app on your site, you can build a chatbot that can integrate with it instead of developing a similar tool from scratch. Remember to think ahead and consider the scalability of your infrastructure as you develop your strategy.

- **Look at data to measure performance**
- Collect data and [customer feedback](#) to evaluate how the bot is performing. The bot itself can capture customer information and analyze how individual responses perform across the entire conversation. This will show you what customers like about AI interactions, help you identify areas of improvement, or allow you to determine if the bot isn't a good fit.

Conversational AI best practices

- Follow these best practices to get the most out of your conversational AI.
- **Create an easy handoff from bot to agent**
- When a conversation requires a human touch or the customer no longer wants to interact with AI, make it easy for the customer to connect with a live agent. The bot will also pass along information the customer already provided, such as their name and issue type.
- “Agents get all that context right away, so they never have to ask customers to repeat themselves,” Lalonde says. “Everybody hates having to repeat themselves, but that goes away in a world where you have conversational AI embedded into the customer experience.”
- **Meet the customer on their preferred channels**
- As customers connect with you over their favorite communication channels, it’s important to have an AI chatbot to meet them where they are. Channels like [social platforms](#), messaging apps, and ecommerce apps help welcome the customer and provide 24/7 service for a great customer experience.
- **Match your AI’s personality to your brand’s tone**
- Chatbots might be the first interaction a customer has with your brand. Tailor their persona to sync with your brand’s tone and to stay consistent across the board. Customers don’t need a comedy routine during their interaction, but they don’t want to talk to a toaster oven, either. As AI and bots become more natural and human-like, businesses can embrace these advances to create better conversational experiences.
- **Keep up with AI advances**
- Adaptability is a crucial element when incorporating technology into your business strategy. AI is constantly evolving—so the flexibility to pivot and quickly adapt must be built into your plans. Set a healthy budget for AI investments to keep up with your competitors. In our CX Trends Report, we found that 68 percent of business leaders already have plans to increase their investments in AI.
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The future of customer experience is conversational AI

- [Seven out of 10](#) consumers now strongly agree that AI is good for society, while 66 percent give AI a thumbs up for making their lives easier. And 69 percent of customers say they're willing to interact with a bot on simple issues—a 23 percent increase from the previous year.
- Companies often view bots as a cost-saving measure, and they certainly can be. But ideally, conversational AI will enhance the capabilities of your support staff, not replace them. “Don’t think of chatbots as a substitute for humans,” Lalonde cautions. “One of the great things about using conversational AI for customer service is the beauty of having both bots and humans working together towards solving customer problems.”
- Conversational AI makes it easier and faster for customers to get answers to simple questions. At the same time, support agents have fewer tickets to resolve, freeing them up to address the complex questions that chatbots and virtual assistants can’t handle. When companies combine the strengths of AI tools and humans, it leads to a better customer experience—and a better bottom line.
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